

Run 10 | TEA

India

IV + controls

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 3,188 (11.57%) **Case time** 59s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9976). Highest F1: H2O GBM (0.9406). For XGBoost, adding controls changed ROC-AUC by +0.0005 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9976	0.9805	98.55	92.39	95.29	0.9382	<1s
H2O GBM	0.9974	0.9772	98.60	92.42	95.76	0.9406	3s
CatBoost	0.9974	0.9773	98.55	92.39	95.29	0.9382	7s
Logistic Regression	0.9973	0.9771	98.53	92.64	94.82	0.9372	2s
Random Forest	0.9973	0.9784	98.49	92.23	94.98	0.9358	1s
Elastic Net	0.9972	0.9754	98.48	92.34	94.66	0.9349	11s
Ridge	0.9971	0.9750	98.57	92.02	95.92	0.9393	12s
Lasso	0.9971	0.9748	98.51	90.51	97.33	0.9380	10s
Decision Tree	0.9940	0.9196	98.48	91.58	95.60	0.9355	<1s
Naive Bayes	0.9377	0.6814	93.25	68.13	78.18	0.7281	10s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,604.93	53.55
2	Established business owner	483.10	16.12
3	Occupational status	125.96	4.20
4	Knowing entrepreneurs	64.48	2.15
5	Age	63.00	2.10
6	Business angel	61.74	2.06
7	Business exit	48.32	1.61
8	Household size	41.80	1.39

Share is the percentage of total forest impurity importance across all 66 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Safety and security	11.99	0.40
2	Investment environment	11.41	0.38
3	Prosperity index score	10.33	0.34

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 10_obj2_india_iv_cv_TEA. Scores and importance are taken directly from saved CSV outputs.